

Compatible anti-forgetting network for re-indexing free lifelong person re-identification

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ABSTRACT

Re-Indexing Free Lifelong Person Re-Identification (RFL-ReID) focuses on addressing the issues of high storage costs and privacy risks in traditional lifelong person re-identification. Unlike conventional lifelong learning settings, RFL-ReID imposes strict restrictions on the use of historical data during model training. Existing approaches mainly rely on knowledge distillation to tackle this challenge. However, due to inevitable information loss during the distillation process and significant domain shifts across tasks—such as differences in viewpoints, resolutions, illumination conditions, and background contexts—the effectiveness of continuous distillation is severely limited. To address these limitations, we propose a novel network architecture—the Compatible Anti-Forgetting Network (CAF). To compensate for the reduced learning capacity caused by restricted access to historical data, we design a deep feature fusion module, which enables the model to extract richer and more multi-dimensional feature representations within a single learning phase, significantly enhancing its generalization ability under the re-indexing-free setting. Furthermore, to mitigate the negative impact of domain discrepancies on knowledge transfer, we introduce an adaptive network constraint module. This module dynamically optimizes the model update path, effectively regulating parameter changes and ensuring a smooth transition between old and new knowledge, thereby improving overall stability and robustness. Experimental results on multiple lifelong person re-identification benchmark datasets demonstrate that our method achieves substantial improvements over current state-of-the-art approaches, with an average mAP gain of 5.1 % and a Rank-1 accuracy improvement of 5.8 %, validating the superior memory retention and adaptation capabilities of CAF in long-term sequential learning tasks.

1. Introduction

Person re-identification (ReID) aims to accurately locate and match the same individual across image sequences captured by different cameras [1]. In recent years, with the development of deep learning techniques, the performance of ReID models has significantly improved [2]. Depending on task settings and application scenarios, ReID has evolved into several important branches, including multimodal ReID [3, 4], visible-infrared ReID [5–7], adaptive ReID [8,9], and unsupervised ReID [10,11], each addressing practical challenges such as multi-modal data fusion, cross-illumination matching, target domain adaptation, and deployment without labeled data.

Building upon these developments, lifelong person re-identification (L-ReID) has emerged as an emerging research direction, focusing on the model's ability to continuously learn from sequential tasks while retaining knowledge from previous ones. The goal is to achieve good

adaptation to new tasks while preserving performance on previously seen ones [12–15]. Currently, most mainstream methods rely on a "retrieval" strategy—storing or replaying historical data—to mitigate catastrophic forgetting [16–18]. However, this approach comes with practical limitations, including high storage costs [19,20] and privacy concerns [21–23], making it impractical to directly reuse historical data.

To overcome these challenges, Re-Indexing Free Lifelong Person Re-Identification (RFL-ReID) [24] was introduced. It aims to break free from reliance on historical data storage and re-indexing [25,26], exploring a new mechanism for efficient lifelong learning without repeatedly accessing old data. This approach significantly reduces storage requirements and security risks while maintaining system performance [27]. However, due to the lack of access to historical data, RFL-ReID imposes higher demands on the model's ability to acquire and retain knowledge.

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Previous work, such as C^2R [24], proposed a Continuous Compatible Transfer (CCT) network that balances new and old knowledge through a dual-stream architecture, attempting to alleviate domain shifts during lifelong learning. Nevertheless, despite its partial success in adapting to changes in feature distributions across tasks, the CCT mechanism still exhibits several limitations:

First, CCT relies on a knowledge distillation [28] framework to preserve performance on historical tasks. However, discrepancies in expressive capacity between teacher and student models inevitably lead to information loss during distillation, especially in multi-task settings where such losses accumulate over time, ultimately affecting the model's memory of early tasks [29,30] (Fig. 1). Second, although the Knowledge Composition Module (KCM) in CCT introduces learnable prototypes to extract new knowledge, its weight allocation mechanism is relatively static and struggles to adapt to complex and evolving task distributions. Meanwhile, the Feature Mapping Module (FMM) focuses primarily on feature space mapping but lacks effective constraints on parameter updates, making it prone to overwrite previously learned knowledge during new task learning (Fig. 2).

To address the above issues, we propose the Compatible Anti-Forgetting Network (CAF), which improves upon existing methods in two key aspects: First, we design a network self-constraint module that dynamically restricts parameter updates during new task learning, preventing excessive interference with previously acquired knowledge. Second, we construct a deep feature fusion module that enables the model to extract more discriminative and generalizable feature representations in a single learning phase, compensating for the information loss caused by knowledge distillation under the re-indexing-free setting. Compared to existing methods, our design not only reduces dependency on historical data but also achieves dual optimization in both model update mechanisms and feature learning strategies, significantly enhancing the model's stability and generalization capability over long-term learning processes.

The main contributions of this work are summarized as follows:

1. We propose the Compatible Anti-Forgetting Network (CAF), offering a novel solution to the problem of information loss caused by knowledge distillation and domain shift [9,31] in Re-Indexing Free Lifelong Person Re-Identification.

2. We design a network self-constraint module, which, for the first time, incorporates gradient sensitivity into the model update process, dynamically suppressing parameter updates that significantly impact old knowledge, thereby overcoming the issue of uncontrolled parameter drift in traditional methods and greatly improving model stability during continuous learning.

3. We develop a deep feature fusion module that encourages the model to extract more generalizable multi-scale features in a single learning step, addressing the limitations of traditional methods in feature expressiveness and enabling efficient cross-task knowledge retention without relying on historical data replay.

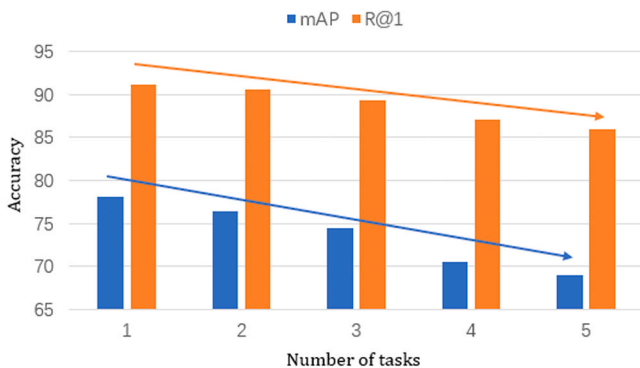


Fig. 1. Changes in mAP and R@1 on market-1501 with the increase in task numbers.

4. Experimental results on multiple benchmark datasets demonstrate that CAF effectively reduces catastrophic forgetting while maintaining strong adaptability to new tasks. The overall performance surpasses current state-of-the-art methods, validating the effectiveness and robustness of the proposed approach.

2. Related work

Re-indexing free lifelong person re-identification: Person re-identification (ReID) has developed several important branches in recent years, including cross-modal ReID [3,4], visible-infrared ReID [5–7], adaptive ReID [8,9], and unsupervised ReID [10,11]. These branches address practical scenarios such as multi-modal data fusion, matching across lighting conditions, target domain adaptation, and deployment without labels. Lifelong person re-identification (Lifelong Person Re-Identification, L-ReID), as a continuous learning task, focuses more on the model's ability to retain old knowledge during continuous learning, particularly addressing catastrophic forgetting. To tackle this issue, researchers have proposed various methods, such as sample replay [17,32] and knowledge distillation [28]. These methods help the model retain old knowledge while learning new tasks by re-indexing old data [25]. Pu et al. [13] first introduced L-ReID and proposed the Adaptive Knowledge Accumulation (AKA [33]) framework, providing new insights into this field. Sun et al. [28] further developed the Patch-KD method, using local features to connect old and new datasets. Xu et al. [34–36] proposed an old knowledge correction mechanism that corrects old model predictions based on new data annotations and enhances knowledge consolidation during new data learning by modeling and rehearsing old domain distributions, without storing any examples. However, when it is impossible to re-index old data, the performance of these methods significantly decreases. To address this problem, Re-Indexing Free Lifelong Person Re-Identification (RFL-ReID) was introduced, aiming to eliminate the reliance on re-indexing old data. Cui et al. [24] proposed a continuous compatible representation method to align the feature spaces of new and old data [37,38] while reducing the impact of catastrophic forgetting. However, due to the strict limitations on data usage in RFL-ReID, continuous information loss during distillation [29,30] and domain shift issues between different tasks [14,31] pose higher challenges for the model.

Fisher information matrix: Fisher information is used to measure the amount of information a single observation provides about an unknown parameter [39]. The Fisher Information Matrix (FIM) describes the curvature near the minimum of the loss surface during the training process of the previous task [40]. Weights that have a greater impact on the model's output show higher values in the FIM, indicating their importance for the previous task [41,42]. Pierre H. Richemond et al. [43] proposed a method to protect the critical knowledge acquired by the model during learning by constraining updates based on Fisher information. This method effectively addresses the catastrophic forgetting problem in continual learning—where models tend to forget previously learned information when learning new tasks. Specifically, it does so by constraining updates to important parameters in the FIM, ensuring that these key parameters are not overly modified when adapting to new tasks [44–46]. This approach retains old knowledge while allowing the model to adapt to new data, thereby improving its performance across multiple tasks. This strategy enables the model to better handle changes between different tasks while maintaining performance on prior tasks.

Depthwise separable convolutions: Depthwise separable convolutions were introduced by Sifre [47], replacing standard convolutions with two simpler steps: Depthwise convolution and pointwise convolution. The introduction of Depthwise separable convolutions aimed to simplify the computation process, reduce model parameters, and thereby enhance computational efficiency and achieve model lightweighting. Previous studies, such as EfficientNet [48] and MobileNet [49,50] have leveraged the lightweight advantages of Depthwise separable convolution models to achieve good results in

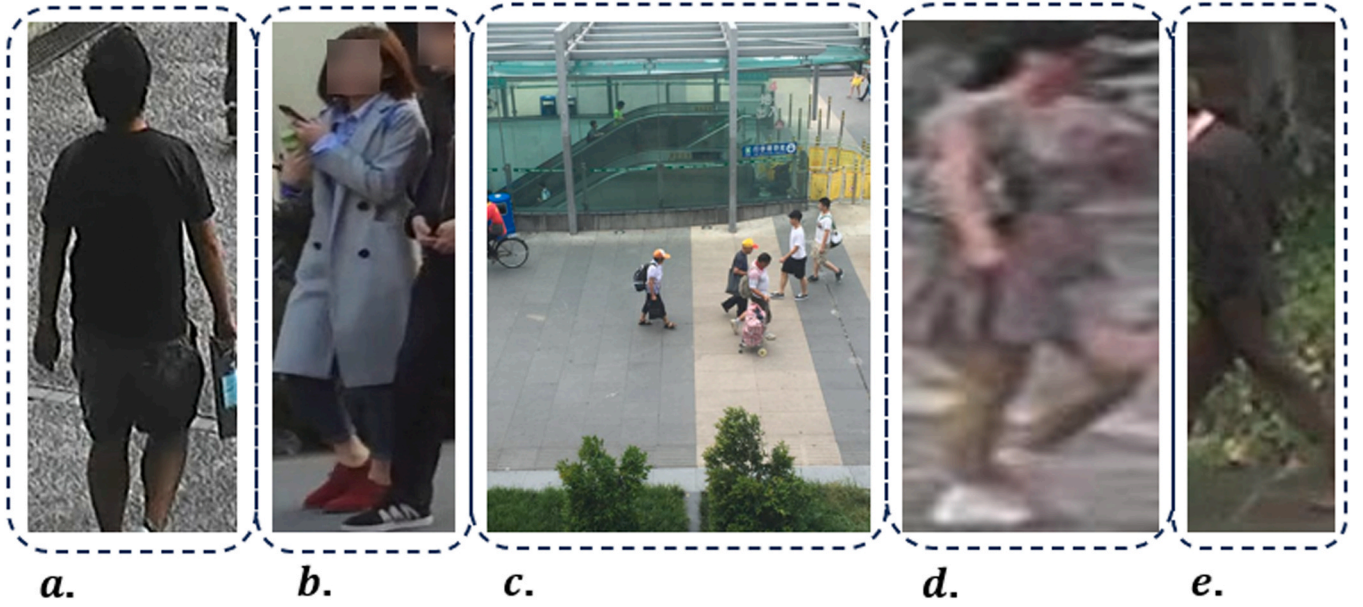


Fig. 2. The great differences between images' data.

resource-constrained environments. Inspired by Inception [51] and Xception [52], we discovered additional benefits of Depthwise separable convolutions in lifelong learning tasks during our experiments. Specifically, each independent channel in Depthwise separable convolutions can be seen as capturing features from different perspectives. By correctly stacking multiple Depthwise separable convolutions with different structures, we can simultaneously acquire richer feature information beyond the original perspective. This is especially effective for re-indexing-free lifelong person re-identification tasks where the use of old data is restricted, significantly enhancing the model's memory capability during single learning sessions.

3. Method

3.1. Network overview

Our CAF network is illustrated in Fig. 3. The primary network is a continuous distillation network designed to continually update model parameters in a re-indexing-free lifelong person re-identification task. Additionally, it includes a Network Self-Constraint (NSC) module to supervise and constrain the distillation process. The NSC module constrains the continuous distillation process through two parts: the model update part based on Fisher information supervision (L_{fim}), and the

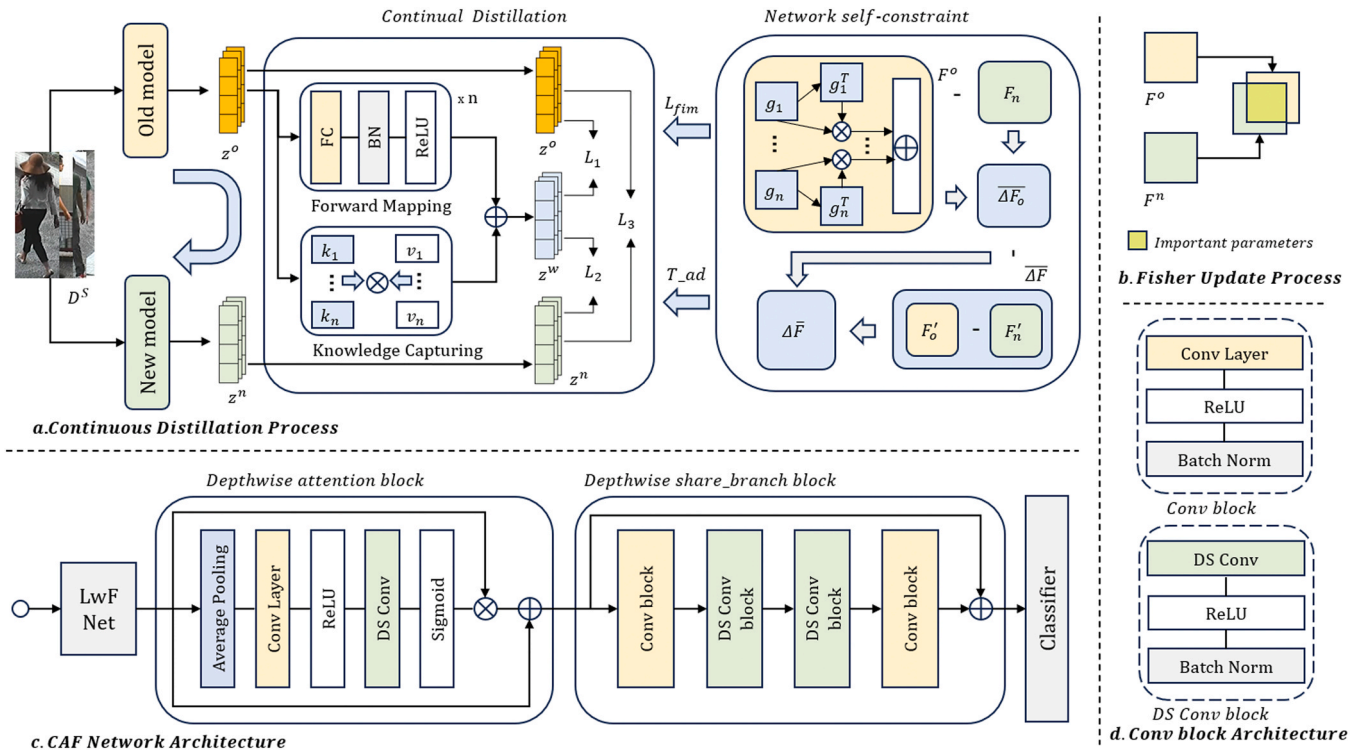


Fig. 3. The architecture of our proposed Compatible Anti-Forgetting Network (CAF) method.

adaptive temperature adjustment part (T_{ad}) based on the variability of input data. In terms of model structure, we have added a deep feature fusion module (DFF) to the traditional LwF [53] network, consisting of two components: a depth attention block and a depth-sharing branch block. These components are used to acquire feature information from images across multiple dimensions, thereby enhancing the model's memory capability.

3.2. Continuous distillation network

Our baseline [24] is a network structure capable of continuously acquiring new knowledge through distillation without replaying old knowledge. Its structure is shown on the left side of Fig. 3a. z^o represents the data features extracted by the old model φ_o from dataset D^s , which are processed through two modules to obtain intermediate features z^w . First, these features pass through a knowledge capture module where the learnable prototype v_o of new knowledge and its capture probability k_o are combined to obtain feature z^t . Second, they pass through a forward mapping module to directly map and obtain feature z^m . Combining z^t and z^m with weights yields the intermediate feature z^w . The old knowledge feature z^o , the intermediate feature z^w , and the new knowledge feature z^n extracted from dataset D^s by the new model φ_n are each subjected to loss calculations, yielding losses L_1 , L_2 , and L_3 , respectively. Their weighted combination forms our baseline loss:

$$L_{base} = L_1 + L_2 + L_3 \quad (1)$$

3.3. Network self-constraint module

To achieve the model's anti-forgetting capability, we were inspired by previous work and focused on the model's Fisher Information Matrix to observe and constrain changes in important parameters. As shown in Fig. 3b, during the model update process, due to the continuous increase of data in lifelong learning tasks, the parts of new and old knowledge learned by the model for different datasets that are more similar tend to have higher generality and are more important. The Fisher Information Matrix measures the degree of influence of model parameters on the loss function. For a parameter p its Fisher information value is defined as:

$$F(p) = E_{x \sim D} \left[\left(\frac{\partial \log P(y|x, \Theta)}{\partial p} \right)^2 \right] \quad (2)$$

Where D is the dataset, x is the input data, y is the corresponding label, and Θ is the model parameter. $P(y|x, \Theta)$ represents the conditional probability distribution of label y given x and Θ . This formula quantifies the importance of each parameter to the model output by calculating the expected squared gradient. However, in practical computation, we usually cannot directly compute the expectation E , so we use sample averaging to approximate the Fisher Information Matrix.

$$F(p) = \frac{1}{N} \sum_{i=1}^N g_i g_i^T \quad (3)$$

where g_i is the gradient vector corresponding to the i -th sample, and N is the total number of samples. To avoid the computational burden caused by large-scale matrix operations, we divide the gradients into multiple small blocks (block), and accumulate the Fisher information block by block. After obtaining the Fisher matrix, we perform singular value decomposition (SVD) to extract the main feature directions, and introduce a nonlinear transformation to compress the singular values, preventing numerical instability. The final weighted Fisher matrix is obtained as:

$$F = U \cdot \text{diag}(S) \cdot V^T, \omega = \sigma \left(\frac{\tilde{S} - \tau}{\alpha} \right) \quad (4)$$

Where S is the diagonal matrix composed of singular values, $\tilde{S} = \log(1+S) + \epsilon$ is a nonlinear transformation of the singular values, ω is the weight term controlled by the Sigmoid function $\sigma(\cdot)$, τ and α are respectively the threshold and slope of the Sigmoid function, ϵ is a small positive number used to prevent numerical issues. Low-rank approximation significantly reduces the storage and computational cost of the Fisher matrix while preserving the main feature information. Based on this, we define a regularization loss:

$$L_{fim} = \frac{1}{2} (\Delta\theta)^T \cdot F \cdot (\Delta\theta) \cdot \omega \quad (5)$$

$\Delta\theta = p - p^o$ represents the difference between the current parameters and the old parameters. This loss term measures the "penalty" of parameter changes under the Fisher matrix, imposing greater constraints on important parameters to effectively alleviate model forgetting when learning new tasks. The overall loss L of our CAF is:

$$L = L_{base} + \mu_1 \cdot L_{fim} \quad (6)$$

In addition, we further utilize the evolution of the Fisher information matrix to adaptively supervise the continual distillation process. By comparing the current and historical Fisher matrices, we calculate an incremental change:

$$\Delta\bar{F} = \bar{\Delta F} - \bar{\Delta F}_o \quad (7)$$

Where $\bar{\Delta F}$ and $\bar{\Delta F}_o$ denote the average Fisher information change of all parameters in the current and previous stages, respectively. The Fisher Information Matrix F represents the current parameter state of the model, and the difference between the two information matrices represents the difference in parameter states, i.e., the amount of change in model parameter information. The increment of Fisher information changes represents the rate of model changes when learning different knowledge, i.e., the impact of different new knowledge on the stability of old knowledge memory.

Since the lifelong learning network undergoes multiple rounds of task learning, the more important knowledge remembered by the model is the commonality of the task data. Therefore, when learning new knowledge with significant differences, the large impact on the model may come from its specificity rather than generality. Thus, we use the increment of Fisher information changes to control the iteration of the continuous distillation network temperature:

$$T(e) = \begin{cases} T(e-1) \cdot (1 + r_T), & \Delta\bar{F} > 0 \\ T(e-1) \cdot (1 - r_T), & \Delta\bar{F} < 0 \\ T(e-1), & \text{otherwise} \end{cases} \quad (8)$$

Where $T(e)$ is the distillation temperature parameter of the model at the e -th epoch, $T(e-1)$ is the distillation temperature of the previous epoch, and r_T is the temperature adjustment rate, controlling the magnitude of temperature adjustments. When the Fisher information change increment $\Delta\bar{F}$ is larger, it indicates that the model has learned more specific knowledge, and the distillation network needs to increase the distillation temperature to maintain memory of old knowledge. Conversely, it indicates that the newly acquired knowledge has better generality, and the network should enhance the acquisition of new knowledge.

3.4. Deep feature fusion module

In addition to using adaptive methods to supervise the continuous iteration and distillation of the model, to address the limitations on the use of old knowledge in no-retrieval tasks, our model enhances the acquisition capability of multi-dimensional new knowledge in single-shot learning. Specifically, we noticed the excellent performance of Depthwise separable convolutions in capturing multi-dimensional features. As shown in Fig. 4, the multiple independent channels of Depthwise separable convolutions can be regarded as multi-channel parallel

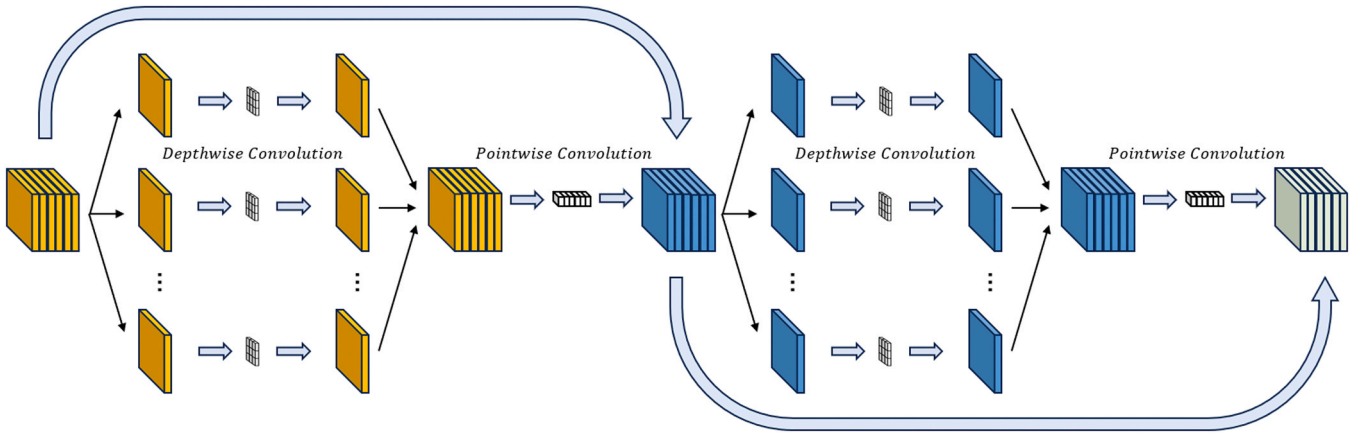


Fig. 4. Deep feature fusion combined structure.

processing from different perspectives. By stacking multiple groups of Depthwise separable convolutions reasonably, more diverse receptive fields can be obtained, enhancing the model's ability to recognize and memorize data features in multiple dimensions.

The structure of our model is shown in Fig. 3c. Based on the LwF [41] net, we designed two block combinations using Depthwise separable convolutions to capture multi-dimensional feature information. First, we designed a deep attention block, which helps the model focus on more important key regions. We first perform global average pooling on the features to extract global information for each channel:

$$y = \Phi(X) \in R^{b \times c \times 1 \times 1} \quad (9)$$

where $\Phi(\cdot)$ represents global average pooling, and $y_{b,c} = \frac{1}{h \cdot w} \sum_{i=1}^h \sum_{j=1}^w X_{b,c,i,j}$. Next, a multi-layer convolution sequence is applied to the globally pooled features to generate channel attention weights through nonlinear transformations. The first convolution layer reduces the number of channels, decreasing the channel count from C to $\frac{C}{\lambda}$, where λ is the channel reduction rate:

$$y_1 = \text{ReLU}(W_1 \cdot y + b_1) \quad (10)$$

where $W_1 \in R^{\frac{C}{\lambda} \times c \times 1 \times 1}$ is the convolution kernel weight, and b_1 is the bias. $\text{ReLU}(x)$ is the activation function, defined as $y_1 = \max(0, y)$. Its purpose is to reduce the number of channels in subsequent depthwise separable convolutions, enabling the model to focus on important features in multi-dimensional vision. The second layer is a Depthwise convolution, which keeps the number of channels unchanged but introduces local receptive fields:

$$y_2 = \text{ReLU}(W_2 \cdot y_1 + b_2) + b_3 \quad (11)$$

W_2 and b_2 are the convolution kernel weights and biases of the depthwise convolution, and W_3 and b_3 are the weights and biases of the pointwise convolution. Next, the feature map after the depthwise separable convolution is up-sampled to match the feature size of the input to the deep attention block:

$$y_3 = \sigma(W_4 \cdot y_2 + b_4) \quad (12)$$

Then, the attention weights y_3 are applied to the input feature map X , and a residual attention method is used to retain the original feature channels, generating the weighted feature map:

$$Z = X + X \cdot \hat{y}_3(X) \quad (13)$$

$\hat{y}_3(X)$ represents extending y_3 to the same shape as X . We improved the attention module by adding multi-dimensional visual capabilities while retaining its ability to extract important information.

Additionally, to enable the model to further acquire and learn feature

knowledge from more perspectives, we designed a deep share_branch block to work in conjunction with the deep attention block. As shown in Figure c, its structure is similar to the deep attention block, but unlike the attention module, the deep share_branch block does not involve pooling. It first passes through a 1×1 Conv block for dimensionality reduction. The Conv block and DS conv block structures are shown in Figure d. If the input feature map size is $H \times W \times C$, the output after the Conv block is:

$$Y_4(x, y, c) = \text{ReLU} \left(\sum_{k=0}^{c-1} W_5(c, k) \cdot X(x, y, k) + b_5(c) \right) \quad (14)$$

where $X(x, y, k)$ is the value of the input feature map at spatial position k and channel c , $W_5(c, k)$ is the weight between the c -th output channel and the k -th input channel, and $b_5(c)$ is the bias term. The output feature map size is $H \times W \times \frac{C}{\lambda}$. Subsequently, it passes through the Depthwise convolution (19) and pointwise convolution (20) in the Depthwise separable convolution:

$$Y_{5d}(x, y, c) = \text{ReLU} \left(\sum_{i=1}^1 \sum_{j=1}^1 W_{6d}(i, j, c) \cdot Y_4(x + i, y + j, c) + b_{6d}(c) \right) \quad (15)$$

$$Y_{5p}(x, y, c) = \text{ReLU} \left(\sum_{k=0}^{c-1} W_{6p}(c, k) \cdot Y_{5d}(x, y, c) + b_{6p}(c) \right) \quad (16)$$

The subsequent Conv block and DS conv block processes are consistent with the above. We denote the outputs of the four convolution blocks as Y_4 , Y_5 , Y_6 and Y_7 . Where Y_4 and Y_7 are responsible for dimensionality reduction and expansion, respectively, while Y_5 and Y_6 maintain the number of channels. We also use residual connections to retain the original feature channels, resulting in the output feature map of the deep share_branch block:

$$Z = X + Y_7(Y_6(Y_5(Y_4(X)))) \quad (17)$$

which we simplify to:

$$Z = X + \tilde{Y}_7(X) \quad (18)$$

After passing through the entire deep fusion module, the feature output can be expressed as:

$$Z = X + X \cdot \hat{y}_3(X) + \tilde{Y}_7(X + X \cdot \hat{y}_3(X)) \quad (19)$$

4. Experiments

4.1. Experimental setup

Datasets: To simulate the complexity of real-world lifelong person re-identification tasks, we conducted extensive experiments on five benchmark person re-identification datasets. These include: Market-1501 [54], CUHK-SYSU [55], DukeMTMC-ReID [56], MSMT17-V2 [57], and CUHK03 [58]. We crop the images of real-world scenes along the boundaries of the persons and reorganize them into a ReID dataset containing the corresponding person information [32]. Meanwhile, to simulate the uncertainty in the process of learning pedestrian identities in real-world scenarios, we used two sequences to train and evaluate the model sequentially [4]. The sequences are as follows: order 1: Market-1501 $\rightarrow$ CUHK-SYSU $\rightarrow$ DukeMTMC-ReID $\rightarrow$ MSMT17-V2 $\rightarrow$ CUHK03; And order 2: DukeMTMC-ReID $\rightarrow$ MSMT17-V2 $\rightarrow$ Market-1501 $\rightarrow$ CUHK-SYSU $\rightarrow$ CUHK03.

Evaluation metrics: To comprehensively evaluate our model's performance, we adopted two main metrics: mean Average Precision (mAP) and Rank@1 accuracy (R@1). By conducting continuous experiments across multiple datasets and evaluating the model on each dataset separately, we averaged all evaluation results to measure the model's overall performance. We compared the model's performance in two scenarios: no-replay lifelong person re-identification tasks and

traditional lifelong person re-identification tasks.

Implementation details. Our proposed CAF method was implemented using PyTorch on an NVIDIA 3090 GPU. We used a ResNet-50 network pretrained on ImageNet as the backbone. During training, we used Adam as the optimizer. Each dataset was trained for 50 epochs, with each epoch containing 150 iterations. The initial learning rate was set to 3.5×10^{-4} , and it was decayed to 1/10 of its original value at the 25th and 35th epochs. The network self-constraint module was introduced starting from the second dataset, with the hyperparameter μ_1 set to 50 and the temperature update rate r_T set to 2 %. The center position for the EWC component was set to 100, and the slope λ was set to 10. Additionally, the channel reduction ratio λ for the deep feature fusion module was set to 8. For image input, we resized images to a uniform resolution of 256×128 and set the batch size to 64, with four images per identity. To ensure fair comparisons, we selected the same number of classes from different datasets.

4.2. Comparison with state-of-the-art methods

In this section, we will delve into our proposed CAF method and conduct a detailed performance comparison with the current state-of-the-art Re-Indexing Free Lifelong Person Re-Identification (RFL-ReID) methods. These methods include *LwF* [53], *CRL* [59], *AKA* [33], *MEGE* [14], *PatchKD* [28], and *C²R* [24]. Our experimental results are

Table 1
Comparison with state-of-the-arts methods.

Task	Method	Market-1501		CUHK-SYSU		DukeMTMC-ReID		MSMT17-V2		CUHK03		Average	
		mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
L-ReID	LwF [41]	56.3	77.1	72.9	75.1	29.6	46.5	6.0	16.6	36.1	37.5	40.2	50.6
	CRL [47]	58.0	78.2	72.5	75.1	28.3	45.2	6.0	15.8	37.4	39.8	40.5	50.8
	AKA [24]	58.1	77.4	72.5	74.8	28.7	45.2	6.1	16.2	38.7	40.4	40.8	50.8
	MEGE [5]	39.0	61.6	73.3	76.6	16.9	30.3	4.6	13.4	36.4	37.1	34.0	43.8
	PatchKD [19]	68.5	85.7	75.6	78.6	33.8	50.4	6.5	17.0	34.1	36.8	43.7	53.7
	C ² R [15]	69.0	86.8	76.7	79.5	33.2	48.6	6.6	17.4	35.6	36.2	44.2	53.7
RFL-ReID	our	67.2	85.6	78.7	80.6	42.5	60.8	10.5	24.9	43.9	44.6	48.6	59.3
	LwF [41]	39.1	58.0	40.0	40.7	7.8	15.3	2.6	7.1	23.3	23.9	22.6	29.0
	AKA [24]	36.1	52.2	38.6	37.6	7.6	13.8	3.1	8.3	26.5	26.5	22.4	27.7
	PatchKD [19]	61.4	78.4	57.8	59.0	20.8	34.4	5.1	12.8	36.0	37.6	36.2	44.4
	C ² R [15]	62.7	79.7	64.4	66.3	26.7	41.7	6.8	15.7	37.2	37.6	39.5	48.2
	our	66.0	82.4	66.0	66.9	34.9	51.2	10.7	23.7	45.2	45.8	44.6	54.0
Task	Method	DukeMTMC-ReID		MSMT17-V2		Market-1501		CUHK-SYSU		CUHK03		Average	
		mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
L-ReID	LwF [41]	42.7	61.7	5.1	14.3	34.4	58.6	69.9	73.0	34.1	34.1	37.2	48.4
	CRL [47]	43.5	63.1	4.8	13.7	35.0	59.8	70.0	72.8	34.5	36.8	37.6	49.2
	AKA [24]	42.2	60.1	5.4	15.1	37.2	59.8	71.2	73.9	36.9	37.9	38.6	49.4
	MEGE [5]	21.6	35.5	3.0	9.3	25.0	49.8	69.9	73.1	34.7	35.1	30.8	40.6
	PatchKD [19]	58.3	74.1	6.4	17.4	43.2	67.4	74.5	76.9	33.7	34.8	43.2	54.1
	C ² R [15]	59.7	75.0	7.3	19.2	42.4	66.5	76.0	77.8	37.8	39.3	44.7	55.6
RFL-ReID	our	57.2	73.2	8.6	21.8	51.3	73.6	78.0	80.0	43.3	45.1	47.7	58.7
	LwF [41]	15.0	22.9	1.2	3.2	9.5	19.4	38.8	37.5	20.2	19.6	16.9	20.5
	AKA [24]	11.1	15.1	1.3	3.2	13.4	27.3	35.9	34.7	25.2	25.6	17.4	21.2
	PatchKD [19]	46.5	60.9	4.0	10.4	31.1	50.5	63.0	64.0	35.8	36.6	36.1	44.5
	C ² R [15]	48.4	63.6	6.2	14.9	37.0	55.6	67.4	68.4	39.2	39.5	39.7	48.4
	our	47.7	64.7	7.4	17.5	46.5	67.2	69.3	70.6	44.8	46.6	43.1	53.3
Task	Method	DukeMTMC-ReID		MSMT17-V2		Market-1501		CUHK-SYSU		CUHK03		Average	
		mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
L-ReID	LwF [41]	42.7	61.7	5.1	14.3	34.4	58.6	69.9	73.0	34.1	34.1	37.2	48.4
	CRL [47]	43.5	63.1	4.8	13.7	35.0	59.8	70.0	72.8	34.5	36.8	37.6	49.2
	AKA [24]	42.2	60.1	5.4	15.1	37.2	59.8	71.2	73.9	36.9	37.9	38.6	49.4
	MEGE [5]	21.6	35.5	3.0	9.3	25.0	49.8	69.9	73.1	34.7	35.1	30.8	40.6
	PatchKD [19]	58.3	74.1	6.4	17.4	43.2	67.4	74.5	76.9	33.7	34.8	43.2	54.1
	C ² R [15]	59.7	75.0	7.3	19.2	42.4	66.5	76.0	77.8	37.8	39.3	44.7	55.6
RFL-ReID	our	57.2	73.2	8.6	21.8	51.3	73.6	78.0	80.0	43.3	45.1	47.7	58.7
	LwF [41]	15.0	22.9	1.2	3.2	9.5	19.4	38.8	37.5	20.2	19.6	16.9	20.5
	AKA [24]	11.1	15.1	1.3	3.2	13.4	27.3	35.9	34.7	25.2	25.6	17.4	21.2
	PatchKD [19]	46.5	60.9	4.0	10.4	31.1	50.5	63.0	64.0	35.8	36.6	36.1	44.5
	C ² R [15]	48.4	63.6	6.2	14.9	37.0	55.6	67.4	68.4	39.2	39.5	39.7	48.4
	our	47.7	64.7	7.4	17.5	46.5	67.2	69.3	70.6	44.8	46.6	43.1	53.3

Order-1: Market-1501 $\rightarrow$ CUHK-SYSU $\rightarrow$ DukeMTMC-ReID $\rightarrow$ MSMT17-V2 $\rightarrow$ CUHK03.

Order-2: DukeMTMC-ReID $\rightarrow$ MSMT17-V2 $\rightarrow$ Market-1501 $\rightarrow$ CUHK-SYSU $\rightarrow$ CUHK03.

summarized in Table 1, which clearly highlights the performance differences among these methods.

RFL-ReID: For the RFL-ReID task, the CAF method demonstrates exceptional performance under two different sequence settings. Specifically, under these two settings, the average Mean Average Precision (mAP) achieved by CAF reached 44.6 % and 43.1 %, while the Rank-1 Accuracy (R@1) was 54.0 % and 53.3 %, respectively. Compared to the current best-performing method, C²R, CAF improved the average mAP by 5.1 % and 3.4 %, and the Rank-1 Accuracy by 5.8 % and 4.9 %, respectively. This significant progress demonstrates that the CAF method can comprehensively capture and memorize target information under strict data usage constraints. It not only effectively retains old knowledge but also maximally absorbs new knowledge, achieving continuous performance improvement without revisiting historical data.

L-ReID: Additionally, the CAF method also performs exceptionally well in traditional lifelong person re-identification tasks. Under the same two sequence settings, the average mAP achieved by CAF was 48.6 % and 47.7 %, while the Rank-1 Accuracy reached 59.3 % and 58.7 %, respectively. Compared to its closest competitor, C2R, CAF improved the average mAP by 4.4 % and 3.0 %, and the Rank-1 Accuracy by 5.6 % and 3.1 %, respectively. These results indicate that CAF not only excels in privacy-sensitive scenarios but also exhibits strong competitiveness in broader general application scenarios. Whether handling strictly data-constrained environments or conventional settings, CAF demonstrates remarkable adaptability and efficiency. The consistent performance of the CAF method across different environments further validates its reliability and robustness in practical applications.

4.3. Ablation study

In this section, we conduct ablation studies based on Sequence 1 to validate the effectiveness of each module in CAF. Our baseline is described in Section 3.2.

4.3.1. Effectiveness of the network self-constraint module

We evaluated the roles of the two sub-modules within the network self-constraint module, as shown in Table 2. Compared to the baseline method, the two modules improved the average mAP and R@1 accuracy for the no-replay lifelong person re-identification task and the traditional lifelong person re-identification task by 0.5 %, 0.4 %, 1.0 %, 1.3 % and 0.9 %, 1.1 %, 0.2 %, 1.6 %, respectively. The complete network self-constraint module improved the two tasks by 1.7 %, 2.1 % and 1.9 %, 1.8 %, respectively. These results show that self-constraint during network updates significantly enhances the model's knowledge retention and anti-forgetting capabilities, benefiting both single-task and multi-task scenarios, as well as privacy-sensitive and general scenarios.

4.3.2. Effectiveness of the deep feature fusion module

Similarly, we further evaluated the role of the deep feature fusion module. As shown in Table 2, compared to the baseline, this module

Table 2
ablation studies of each component in CAF on Order-1.

Base	NSC		DFF	L-ReID		RFL-ReID	
	L_{fim}	T_{ad}		mAP	R@1	mAP	R@1
✓	-	-	-	44.2	53.7	39.5	48.2
✓	✓	-	-	44.7	54.1	40.5	49.5
✓	-	✓	-	45.1	54.8	39.7	49.8
✓	✓	✓	-	45.9	55.8	41.4	50.0
✓	-	-	✓	47.7	58.1	42.7	52.0
✓	✓	-	✓ (without DS)	46.7	57.0	41.4	50.5
✓	✓	✓	✓	48.6	59.3	44.6	54.0

(The second-to-last row indicates the results of removing Depthwise separable convolution from our original CAF method.)

improved the average mAP and R@1 accuracy for the two lifelong person re-identification tasks by 2.4 %, 2.9 % and 2.5 %, 3.5 %, respectively. These results demonstrate that the deep feature fusion module effectively enhances memory retention for new knowledge and preserves old knowledge, leading to significant improvements in overall performance.

4.3.3. Effectiveness of multi-perspective feature fusion

To verify the role of Depthwise separable convolutions in capturing multi-perspective features, we replaced them with standard convolutions and designed two components to replace the deep attention block and the deep share branch block, as shown in Fig. 5. As shown in Table 2, We observed that Depthwise separable convolutions play a significant role in acquiring and fusing multi-dimensional features in our CAF network, greatly enhancing the model's ability to acquire and memorize new knowledge.

4.4. Hyperparameter analysis

In this study, we conduct a systematic analysis of several key hyperparameters in the proposed CAF method to evaluate their impact on model performance.

Specifically, As shown in Table 3, the loss coefficient μ_1 directly influences the model's ability to retain previously learned knowledge during the knowledge updating process. If μ_1 is set too high, it overly constrains weight updates, thereby limiting the model's capacity to adapt to new tasks. Conversely, if μ_1 is too low, the model fails to effectively prevent catastrophic forgetting. Based on a comprehensive evaluation of both mAP and R@1 performance, we set $\mu_1 = 50$, which achieves a favorable trade-off between retaining old knowledge and acquiring new knowledge.

Moreover, the temperature scaling rate δ_T also significantly affects the model's memory retention capability. An excessively large or small δ_T may lead to suboptimal knowledge preservation and reduced generalization across datasets. Therefore, we determine that setting δ_T to 2 % strikes an appropriate balance between maintaining knowledge stability and enhancing model flexibility.

Regarding the design of the deep feature fusion module, we experimentally investigate the channel reduction ratio λ at both the early and later stages of the network. The optimal value of λ is determined to be 8. Experimental results show that an overly large λ restricts the model's ability to capture multi-dimensional and multi-perspective feature information, while an overly small λ increases computational overhead and may neglect important feature representations.

Finally, we also analyze the impact of batch size on overall model

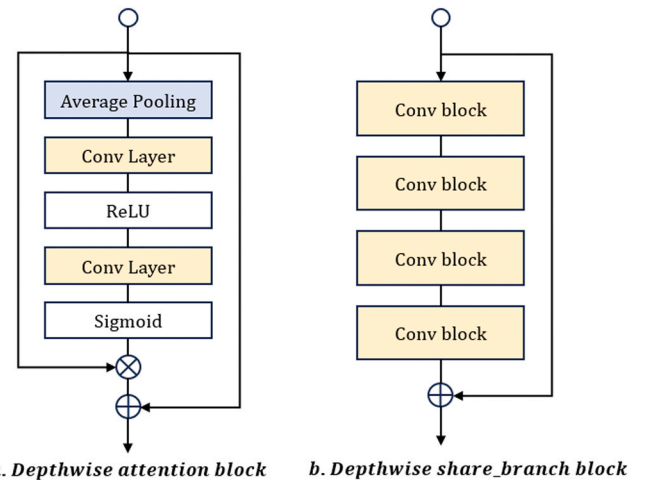


Fig. 5. Deep Feature Fusion Module without Depthwise separable convolutions.

Table 3

Hyperparameter analysis in CAF on order-1.

	Value of μ_1				
	1	10	50	100	500
mAP	39.6	40.0	40.5	40.1	39.1
R@1	48.5	49.2	49.5	49.6	47.6
Value of r_T					
	0.5 %	1 %	2 %	3 %	5 %
mAP	39.0	39.9	39.7	39.1	38.5
R@1	48.3	49.2	49.8	48.4	48.0
Value of λ					
	1	4	8	16	32
mAP	40.5	41.1	42.0	41.7	40.0
R@1	50.0	51.3	51.7	49.2	49.3
Value of Batch size					
	16	32	64	128	256
mAP	44.1	44.2	44.6	44.1	41.2
R@1	53.8	53.5	54.0	53.9	49.8

(All measurement values have a standard deviation of ± 0.15 .)

performance. Experimental results indicate that a large batch size tends to cause overfitting due to the homogeneity of training samples within each update, thereby degrading generalization on unseen data. On the other hand, a small batch size leads to high variance in gradient estimation, resulting in unstable parameter updates and slower convergence. After careful consideration of both training efficiency and model performance, we choose a batch size of 64 as the optimal setting.

In summary, appropriate hyperparameter settings are essential for ensuring the effectiveness and reliability of the CAF method in the context of incremental knowledge learning.

4.5. Stability analysis

To evaluate the stability and robustness of our method under different task sequences, we extend the two commonly used dataset ordering protocols in person re-identification with three additional task sequences, defined as follows: Order3: MSMT17-V2 $\rightarrow$ Market-1501 $\rightarrow$ CUHK-SYSU $\rightarrow$ DukeMTMC-ReID $\rightarrow$ CUHK03; Order4: MSMT17-V2 $\rightarrow$ Market-1501 $\rightarrow$ DukeMTMC-ReID $\rightarrow$ CUHK-SYSU $\rightarrow$ CUHK03; Order5: DukeMTMC-ReID $\rightarrow$ Market-1501 $\rightarrow$ MSMT17-V2 $\rightarrow$ CUHK-SYSU $\rightarrow$ CUHK03.

The experimental results are summarized in Table 4. It can be observed that our method achieves consistently strong performance across all five task sequences, with minimal performance fluctuations between stages. This demonstrates that the proposed approach exhibits good stability and generalization capability when faced with diverse task evolution paths.

Furthermore, to assess the model's ability to retain performance over an increasing number of tasks—as well as its effectiveness in mitigating information loss during continuous distillation—we conduct experiments on two different task sequences. Specifically, we evaluate how model performance on historical datasets evolves as new tasks are introduced. For this purpose, we compute the cumulative performance drop across all datasets relative to their initial performance, which serves as a metric for evaluating the model's resistance to catastrophic forgetting.

Table 4

Performance of CAF under different task orders.

	L-ReID		RFL-ReID	
	mAP	R@1	mAP	R@1
Order1	48.6	59.3	44.6	54.0
Order2	47.7	58.7	43.1	53.3
Order3	48.2	62.0	43.4	55.6
Order4	48.1	62.0	43.5	55.6
Order5	47.6	58.3	43.0	55.3

As shown in Table 5, compared to the baseline method, our approach significantly reduces the degree of knowledge forgetting under both task sequences, demonstrating a superior ability to maintain task-level performance and effectively alleviating the issue of information loss during continual distillation.

In summary, the above experiments fully validate the stability and robustness of our method under varying task orders, as well as its effectiveness in preserving learned knowledge throughout long-term continual learning. These results further confirm the superiority of the proposed method in incremental learning scenarios.

4.6. Temperature adaptation analysis

To gain deeper insight into the behavior of the key component NSC in the proposed method during training, we record and visualize the adaptive adjustment of the distillation temperature T in response to changes in the Fisher information matrix throughout the continual learning process. Specifically, during the learning of the 2nd to 5th tasks, we sample the temperature value every 10 training epochs and plot its evolution over time. The first task is excluded from this analysis as no knowledge distillation is applied at that stage.

As shown in Fig. 6, the distillation temperature rises rapidly at the early stages of each new task. This phenomenon is primarily caused by the model adapting to the new data distribution and task-specific characteristics, which leads to a significant change in the Fisher information matrix. The increased Fisher variation triggers the adaptive temperature-updating mechanism within NSC, resulting in a higher temperature that enhances the protection of old knowledge and helps prevent catastrophic forgetting.

As training progresses, the model becomes better adapted to the current task, and learning stabilizes. Consequently, the variations in the Fisher information matrix decrease, and the distillation temperature gradually declines and converges to a lower value. This dynamic behavior reflects the model's ability to regulate learning intensity based on the importance of historical parameters, achieving an effective balance between knowledge retention and the acquisition of new information.

In summary, this mechanism not only demonstrates the effectiveness of the NSC module in mitigating catastrophic forgetting, but also reveals, from the perspective of training dynamics, how the Fisher-based regularization loss and the adaptive temperature strategy work in coordination during model updates. This contributes to the overall stability and generalization capability of the model in continual learning scenarios.

4.7. Lightweight analysis

To evaluate the resource consumption of our method in practical deployment scenarios and explore its applicability in environments with limited computational resources, we conducted systematic experiments to measure the model's runtime, memory usage, and computational complexity—including the number of parameters and floating-point operations (FLOPs). We compared our method with the baseline approach, and the results are summarized in the Table 6.

Experimental results show that our method achieves significant performance improvement while introducing only a minimal amount of additional computational overhead. Compared to the baseline method, the increases in both model parameters and computational cost remain at a relatively low level, yet the performance gain is much more substantial, demonstrating a favorable trade-off between effectiveness and efficiency. Specifically, in our design, we extensively employ depthwise separable convolutions to replace traditional convolution modules. This allows us to retain strong multi-dimensional feature extraction capability while effectively reducing computational burden and improving inference efficiency.

In summary, the proposed method not only exhibits clear advantages

Table 5
Task-level performance.

Method	Datasets	Market-1501		CUHK-SYSU		DukeMTMC-ReID		MSMT17-V2		CUHK03		Sum	
		mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
Our	1	73.4	89.1	-	-	-	-	-	-	-	-	-	-
	2	72.7	88.4	82.5	84.5	-	-	-	-	-	-	-	-
	3	70.1	87.7	81.4	82.4	47.6	64.0	-	-	-	-	-	-
	4	68.5	86.8	79.9	81.0	45.0	62.5	13.9	31.9	-	-	-	-
	5	67.2	85.6	78.7	80.6	42.5	60.8	10.5	24.9	43.9	44.6	-	-
	D-value	6.2	3.5	3.8	3.9	5.1	3.2	3.4	7.0	-	-	18.5	17.6
Baseline	1	78.1	91.2	-	-	-	-	-	-	-	-	-	-
	2	78.0	90.6	84.1	86.7	-	-	-	-	-	-	-	-
	3	74.5	89.4	80.2	84.0	49.5	65.6	-	-	-	-	-	-
	4	69.9	87.1	78.5	81.5	41.8	60.2	12.2	27.9	-	-	-	-
	5	69.0	86.8	76.7	79.5	33.2	48.6	6.6	17.4	35.6	36.2	-	-
	D-value	9.1	4.4	7.4	7.2	16.3	17.0	5.6	10.5	-	-	38.4	39.1

Method	Datasets	DukeMTMC-ReID		MSMT17-V2		Market-1501		CUHK-SYSU		CUHK03		Sum	
		mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
Our	1	67.1	82.3	-	-	-	-	-	-	-	-	-	-
	2	66.8	81.2	12.1	29.4	-	-	-	-	-	-	-	-
	3	65.4	80.4	10.8	27.3	59.6	80.8	-	-	-	-	-	-
	4	63.5	77.9	10.5	26.0	57.3	78.8	81.0	82.9	-	-	-	-
	5	57.2	73.2	8.6	21.8	51.3	73.6	78.0	80.0	43.3	45.1	-	-
	D-value	9.9	9.1	3.5	7.6	8.3	7.2	3.0	2.9	-	-	24.7	26.8
Baseline	1	70.0	85.1	-	-	-	-	-	-	-	-	-	-
	2	67.9	82.5	9.8	26.1	-	-	-	-	-	-	-	-
	3	65.4	80.3	7.7	20.9	52.2	75.5	-	-	-	-	-	-
	4	64.4	79.7	7.8	20.4	45.4	69.3	77.3	79.4	-	-	-	-
	5	58.1	74.4	6.3	17.2	39.3	64.0	74.6	77.4	36.2	37.6	-	-
	D-value	11.9	10.7	3.5	8.8	12.9	11.5	2.7	2.0	-	-	31	33

Order-1: Market-1501→CUHK-SYSU→DukeMTMC-ReID→MSMT17-V2→CUHK03.

Order-2: DukeMTMC-ReID→MSMT17-V2→Market-1501→CUHK-SYSU→CUHK03.

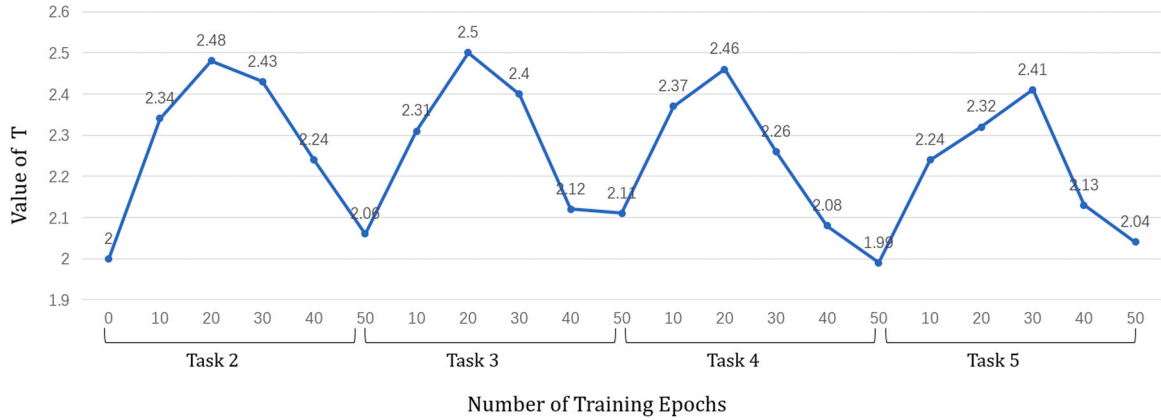


Fig. 6. Evolution of distillation temperature T with training epochs.

Table 6
comparison of computational overhead.

	Runtime	memory usage	Parameters	FLOPs
Our	3 h36 min	9517.2 M	25.8 M	271.4 G
Baseline	3 h15 min	9229.9 M	23.5 M	261.5 G

in terms of performance, but also shows strong competitiveness in model lightweighting and resource utilization efficiency. It therefore holds great potential for deployment in edge devices or low-power applications.

4.8. Visualization

To visually validate the effectiveness of our proposed Compatible Anti-Forgetting Network (CAF) method, we used Grad-CAM [60] to compare CAF with the baseline method. This visualization clearly

demonstrates the superiority of the deep feature fusion module in CAF and its specific contributions to improving model performance. As shown in Fig. 7, after applying our CAF method, we observed a significant increase in attention to the original features compared to the baseline method, specifically through enhanced attention mechanisms across multiple dimensions. This means the model can capture and learn feature information from multiple perspectives, greatly improving its learning capability.

Enhanced attention mechanism: Through the deep feature fusion module, the model focuses not only on the most prominent parts of the image but also identifies and utilizes less conspicuous yet equally important features for classification tasks. This enables the model to memorize more complex and richer feature structures during single-shot training.

Multi-perspective feature learning: By promoting effective integration of features at different levels, the deep feature fusion module allows the model to learn more diverse and comprehensive information

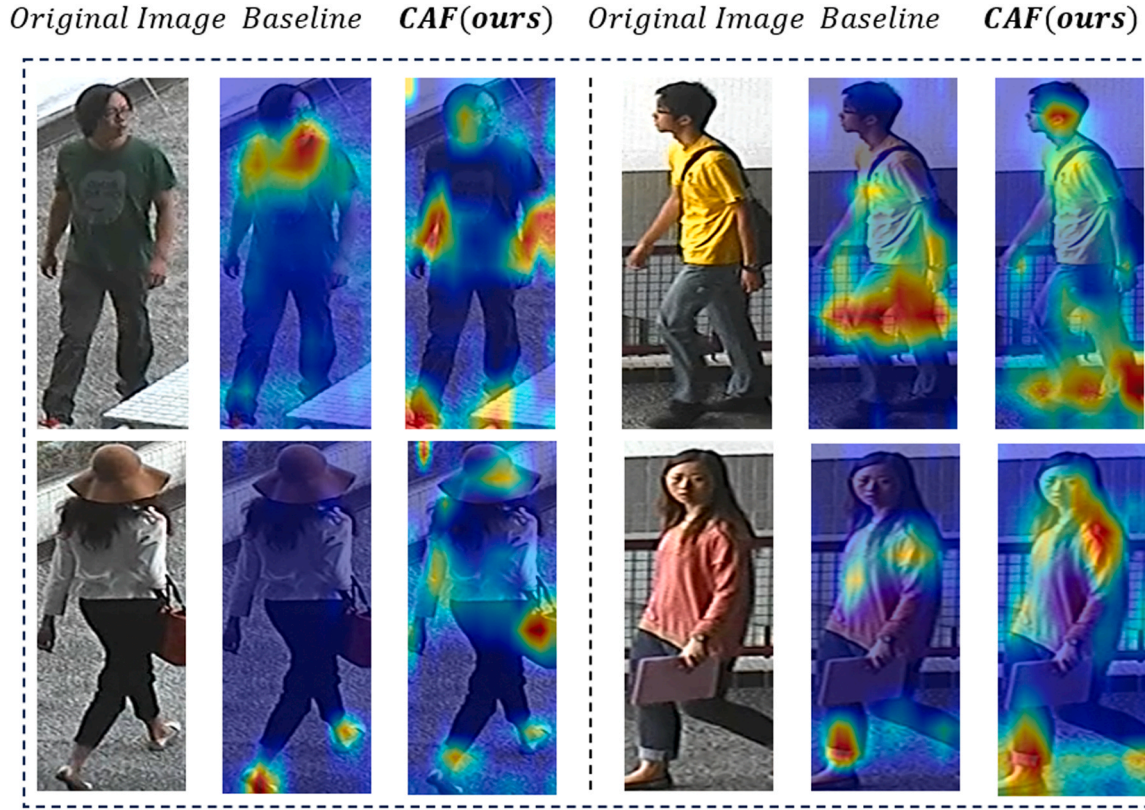


Fig. 7. Grad-CAM results of our CAF compared with Base method.

from input data. This is particularly important for person re-identification tasks because, in real-world scenarios, a pedestrian's appearance may vary due to changes in perspective, occlusion, etc.

These improvements are especially applicable to RFL-ReID task, where data usage is restricted. The newly learned multi-dimensional features help the model memorize more complex structures in single-shot training, thereby enhancing its performance in RFL-ReID tasks.

5. Limitation

Although our method CAF significantly enhances the model's learning capability and achieves excellent performance in both RFL-ReID and traditional L-ReID tasks, it still has some limitations. First, the superior performance of CAF relies on deep learning from training data, which means that if the training data contains significant noise, it could substantially degrade CAF's performance. Additionally, while CAF performs well in RFL-ReID tasks, its generalization ability in other types of lifelong learning tasks needs further validation. In future work, we will focus on exploring the potential of CAF in different lifelong learning scenarios and attempt to use previously learned knowledge for selective transfer on potentially noisy data to improve the model's robustness and generalization.

6. Conclusion

In this paper, we propose a novel network, named CAF, to address the problem of Re-Indexing Free Lifelong Person Re-Identification (RFL-ReID). To overcome the limitations in data usage under this setting, we design a deep feature fusion module, which enables the model to capture multi-dimensional feature information during single-stage learning, thereby enhancing its representational capability in the absence of data replay. Moreover, to adapt to domain shifts across different datasets and reduce their negative impact on distillation performance, we introduce a network self-constraint module that dynamically and adaptively

regulates the model updating and knowledge distillation process. This mechanism ensures a balanced trade-off between retaining old knowledge and acquiring new knowledge, thus improving the overall stability and generalization ability of the model. We conduct extensive experiments on multiple lifelong person re-identification benchmarks. Compared with state-of-the-art methods, our approach achieves an average mAP improvement of 5.1 % and a 5.8 % increase in R@1 accuracy, demonstrating the effectiveness and superiority of our method.

CRedit authorship contribution statement

Boxu Ren: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yongping Xie:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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